

Data Visualization Project on Playable History (Historical Video Games)

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Project Overview

This project began from a question that came out of my broader coursework this quarter. In my classes on “Interactive Systems” and “A Critical Approach to Sid Meier’s Civilization”, I kept returning to the idea that games do not simply show worlds or histories to players. They organize worlds into systems. A historical game is therefore not just a representation of the past; it also defines what the player can do with the past. My dashboard, **Playable History**, explores this idea through a curated corpus of historical video games. Instead of asking whether games are historically “accurate,” the project asks what kind of historical agency they make playable. I organize the corpus around two interpretive poles: **command/control history** and **lived/experiential history**. Command/control history describes games where the player acts through warfare, strategy, empire, management, politics, and diplomacy. Lived/experiential history describes games where the player encounters history through society, culture, narrative, place, open-world exploration, or embodied experience.

The project is inspired by historical game studies and especially by quantitative approaches to historical video games. Yannick Rochat’s work on historical video games uses metadata to study large-scale patterns across historical games rather than relying only on close readings of individual titles.¹ My project works at a much smaller scale, but it follows a similar “distant reading” impulse: using metadata to ask what patterns become visible across a corpus of games.

Data Source and Cleaning Process

My original plan was to collect data through the MobyGames API. However, when I tried to access MobyGames programmatically, I ran into access limits/paywall issues that made a full API-based approach unrealistic within the time and scope of this project. Instead, I built a curated dataset manually. The final dataset contains **44 historical video games** selected because they explicitly engage historical settings, historical periods, warfare, empire, political systems, social conditions, or historically inspired worlds.

The dataset uses metadata and theme information collected from MobyGames pages, then cleaned and reorganized into a form that supports my research question. The final dashboard uses

¹ Rochat, Yannick. “A Quantitative Study of Historical Video Games (1981–2015).” In *Historia Ludens: The Playing Historian*, edited by Alexander von Lünen, Katherine J. Lewis, Benjamin Litherland, and Pat Cullum, 3–19. New York: Routledge, 2020.

`cleaned_games_playable_history.csv`, which includes fields such as title, release year, main genre, rating, command/control score, lived/experiential score, historical play type, and interpretive tag columns such as `has_warfare`, `has_strategy_system`, `has_society_culture`, and `has_narrative_adventure`.

The most important data cleaning step was not just removing errors, but translating messy game metadata into interpretive categories. I grouped relevant tags into two main scores. The **command/control score** counts warfare, strategy/system, management/economy, empire/expansion, and politics/diplomacy. The **lived/experiential score** counts society/culture, narrative/adventure, and open world/sandbox. These scores are not objective measures of historical value. They are analytical categories I created to test my hypothesis visually.

Visualization Choices and Motivation

The dashboard uses three main visualizations and one smaller reception note.

The first and most important visualization is the **Playable History Grid**. It places games according to their command/control score and lived/experiential score. Because these scores are small integers, I decided not to use a normal scatterplot; too many games would overlap at the same coordinates. Instead, the final version uses a bubble grid, where each bubble represents a score combination and bubble size shows how many games fall into that position. Hover text reveals the individual titles. This visualization is meant to serve as the conceptual map of the project: it shows whether the corpus clusters around command history, lived history, hybrid history, or mixed/low-signal cases. This graph also includes an optional release-year animation showing how the corpus accumulates over time, which can be more interesting if the data base was larger and across larger time spans.

The second visualization is **Historical Game Themes: Experiential vs Command**. This is a diverging bar chart. Command/control modes extend in one direction, while lived/experiential modes extend in the opposite direction. I chose this design because a normal bar chart made the categories feel like a simple list. The diverging form better expresses the project's central contrast: history as something to command versus history as something to inhabit. The chart also includes a percentage/count toggle, so the user can switch between relative and absolute views.

The third visualization is the **Genre Orientation Plot**. Earlier versions of the project used a heatmap, but that did not work well because the dataset is small and categorical. The heatmap created harsh blocks and made the interpretation less clear. The final version uses a dumbbell chart instead. For each genre, it compares the average command/control score with the average lived/experiential score. The dashboard allows sorting, minimum genre-size filtering, and genre highlighting.

Finally, ratings are included only as a **reception clue**, not as the center of the project. The dashboard treats ratings as useful context but not as proof that one type of historical play is better or causes higher scores. This was an important choice because I did not want the project to become a ratings analysis.

Findings, Limits, and Future Directions

Overall, the dashboard suggests that many historical games in my corpus lean toward **command/control history**. Tags related to warfare, strategy, management, empire, and politics appear frequently, which supports my central idea that historical games often make the past playable through systems of control, optimization, and decision-making. However, the project also shows that this is not the only way games represent history. Some games are better understood as **lived/experiential history**, foregrounding narrative, social experience, survival, place, or embodied perspective. Hybrid cases are especially interesting because they combine both forms: they let players act within large systems while also inhabiting historical worlds more personally.

At the same time, I do not want to overstate the findings. This is a **curated exploratory corpus** of 44 games, not a random or comprehensive sample of all historical video games. Because I selected the games manually, the dataset likely overrepresents well-known titles and franchises. The tags are also interpretive: another researcher might classify some games differently. MobyGames was useful as a metadata source, but its categories were not designed for my exact research question, so I had to reorganize the data into my own analytical framework. Ratings are also included only as a reception clue, not as proof of quality or causation.

If I continued this project, I would build a much larger dataset through API access or a more systematic collection process. I would also compare historical games with non-historical games to see whether command/control forms are especially common in historical games or common across games more generally.

Collaboration Statement

I used ChatGPT during this project as coding, design, and writing assistants. ChatGPT helped me refine the hypothesis, think through visualization choices, and to edit parts of the writeup. I also used ChatGPT to refine my Streamlit implementation, debugging, and did some code cleanup. The research question, data collection and cleaning, interpretive categories, streamlit code draft, final dashboard decisions, and write up draft are my own. This use of AI supported the process, but it did not replace my role in designing the argument, writing code draft/pseudo code and making final decisions.